RAJALAKSHMI INSTITUTE OF TECHNOLOGY

(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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SEMESTER IV

AD23411 – DATA ANALYTICS LABORATORY

MINI PROJECT REPORT

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PROJECT TITLE	**Develop a transport route optimization analytics project using travel and traffic datasets**
DATE OF SUBMISSION	
FACULTY IN-CHARGE	**Mrs.Bhavani M – Assistant Professor**

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Internal Examiner **External Examiner**

INTRODUCTION

Transport systems play a vital role in daily life, connecting people, goods, and services efficiently. However, increasing traffic congestion, fuel consumption, and travel delays have made route optimization an essential problem in modern cities.

Transport Route Optimization is the process of identifying the most efficient route between two or more locations by considering factors such as distance, travel time, traffic conditions, and road constraints. With the help of data analytics and machine learning, it is possible to analyze large volumes of traffic and travel data to improve route planning.

This project focuses on developing a Transport Route Optimization Analytics System that uses travel and traffic datasets to determine the best possible routes. By analyzing historical traffic data, travel time, distance, and congestion levels, the system provides optimized routes for users.

The project integrates data preprocessing, visualization, and predictive modeling techniques to deliver accurate and efficient route recommendations.

PROBLEM STATEMENT

Urban transportation systems face major challenges due to traffic congestion, inefficient route planning, and increased travel time. Traditional navigation systems often rely on static maps and may not effectively consider real-time traffic conditions or historical patterns.

Drivers and transport services often choose routes based only on distance, which may not always result in the fastest or most efficient path. This leads to increased fuel consumption, delays, and environmental pollution.

There is a need for an intelligent system that can analyze travel and traffic data to determine optimal routes dynamically. The system should consider multiple parameters such as traffic density, road conditions, travel time, and distance to provide better route recommendations.

GOAL

The main goal of this project is to develop a Transport Route Optimization System using data analytics techniques.

Objectives:

- Collect travel and traffic datasets
- Analyze traffic patterns and congestion
- Identify shortest and fastest routes
- Apply optimization algorithms
- Visualize traffic data and routes
- Improve travel efficiency and reduce delays

THEORETICAL BACKGROUND

1. Route Optimization

Route optimization determines the best path between locations based on various constraints.

Types:

- Shortest Path (Distance-based)
- Fastest Path (Time-based)
- Least Congested Path

2. Algorithms Used

- Dijkstra's Algorithm
- A* (A-Star) Algorithm
- Floyd-Warshall Algorithm

3. Data Analytics

Used to analyze:

- Traffic flow
- Travel time
- Road conditions
- Peak hours

4. Data Preprocessing

Includes:

- Cleaning traffic data
- Handling missing values
- Normalizing travel time
- Encoding categorical data

METHODOLOGY

1. Data Collection

Travel and traffic datasets are collected from sources such as GPS logs, Google Maps datasets, or open transport data.

Data Includes:

- Source and Destination
- Distance
- Travel Time
- Traffic Density
- Road Type

2. Data Preprocessing

- Remove missing values
- Standardize data
- Convert categorical features
- Normalize numerical values

3. Exploratory Data Analysis

Analyze:

- Traffic patterns
- Peak congestion times
- Distance vs Travel Time
- Route efficiency

4. Model Implementation

Apply route optimization algorithms:

- Dijkstra Algorithm for shortest path
- A* Algorithm for faster computation

5. Visualization

- Traffic heatmaps
- Route maps
- Travel time graphs

6. Prediction System

System suggests:

- Best route
- Estimated travel time
- Traffic conditions

SAMPLE DATA

Source	Destination	Distance (km)	Travel Time (min)	Traffic Level
A	B	10	20	Medium
A	C	15	25	High
B	C	8	15	Low
C	D	12	30	High
B	D	20	35	Medium

GRAPHS AND CHARTS

1. Traffic Density Analysis

Shows traffic levels across different routes and times of the day. Helps identify peak congestion periods.

2. Distance vs Travel Time Graph

Displays the relationship between distance and actual travel time, highlighting how traffic affects travel duration.

3. Route Efficiency Chart

Compares multiple routes between the same locations to determine the most efficient path.

4. Traffic Heatmap

Represents high and low traffic zones visually for better route planning.

5. Travel Time Distribution

Shows variation in travel time for different routes under different traffic conditions.

IMPLEMENTATION CODE AND DATASET USED

Category	Details
Dataset Name	Traffic and Travel Dataset
Domain	Transportation Analytics
Objective	Optimize routes using traffic data
Data Source	GPS Data / Open Traffic Dataset
Features	Distance, Time, Traffic, Route
Target	Optimal Route
Libraries Used	pandas, numpy, networkx, matplotlib, flask
Algorithms	Dijkstra, A*
Backend	Flask

DATASET DESCRIPTION

The dataset used in this project contains travel and traffic-related data collected from GPS systems, transportation records, or open traffic datasets. It is designed to analyze traffic patterns and optimize routes between different locations.

Each record in the dataset represents a connection between two locations and includes information about distance, travel time, and traffic conditions. This dataset helps in identifying efficient routes and reducing travel delays.

The dataset is stored in tabular format, making it suitable for data analysis and route optimization algorithms such as Dijkstra and A*.

IMPORTANT ATTRIBUTES IN DATASET

- Source Location
- Destination Location
- Distance (in km)
- Travel Time (in minutes)
- Traffic Level (Low / Medium / High)
- Road Type (Highway / City Road / Rural Road)
- Average Speed
- Congestion Level
- Weather Condition (optional)
- Time of Travel (Peak / Off-Peak)

TARGET VARIABLE

The output of the system is:

- Optimal Route Path
- Estimated Travel Time
- Total Distance
- Traffic Condition

PURPOSE OF DATASET

This dataset is used for:

- Identifying the most efficient travel routes
- Analyzing traffic patterns and congestion levels
- Reducing travel time and fuel consumption
- Improving route planning using optimization algorithms
- Supporting transportation systems and smart city solutions

PYTHON CODE

1. MAIN PROJECT STRUCTURE

```
project/
│
├── app.py
├── model.py
├── data.csv
├── templates/
│   ├── index.html
│   ├── result.html
│
├── static/
│   └── style.css
```

2. DATASET (data.csv)

```
source,destination,distance,time,traffic
A,B,10,20,medium
A,C,15,25,high
B,C,8,15,low
C,D,12,30,high


B,D,20,35,medium
A,D,30,50,high
```

3. MODEL FILE (model.py)

```
import pandas as pd
import networkx as nx

class RouteOptimizer:

    def __init__(self, file_path):
        self.df = pd.read_csv(file_path)
        self.graph = nx.Graph()
        self.build_graph()

    def traffic_weight(self, traffic):
        if traffic == "low":
            return 1
        elif traffic == "medium":
            return 2
        else:
            return 3

    def build_graph(self):
        for _, row in self.df.iterrows():
            weight = row['time'] * self.traffic_weight(row['traffic'])
            self.graph.add_edge(
                row['source'],
                row['destination'],
                weight=weight,
                distance=row['distance'],
                time=row['time']
            )

    def find_best_route(self, source, destination):
        path = nx.dijkstra_path(self.graph, source, destination, weight='weight')
        total_time = 0
        total_distance = 0
```

```
        for i in range(len(path) - 1):
            edge = self.graph[path[i]][path[i + 1]]
            total_time += edge['time']
            total_distance += edge['distance']

        return path, total_time, total_distance
```

4. MAIN FLASK APPLICATION (app.py)

```
from flask import Flask, render_template, request
from model import RouteOptimizer

app = Flask(__name__)

optimizer = RouteOptimizer("data.csv")

@app.route('/')
def home():
    return render_template("index.html")


@app.route('/result', methods=['POST'])
def result():
    source = request.form['source']
    destination = request.form['destination']

    try:
        path, time, distance = optimizer.find_best_route(source, destination)


 return render_template(
            "result.html",
            path=" → ".join(path),
            time=time,
            distance=distance
        )
    except:
        return render_template("result.html", error="Route not found!")


if __name__ == "__main__":
    app.run(debug=True)
```

5. FRONTEND (templates/index.html)

```
<!DOCTYPE html>
<html>
<head>
    <title>Route Optimization</title>
    <link rel="stylesheet" href="/static/style.css">
</head>
<body>

<div class="container">
    <h1>Transport Route Optimization</h1>

    <form action="/result" method="POST">
        <label>Source:</label>
        <input type="text" name="source" required>

        <label>Destination:</label>
        <input type="text" name="destination" required>

        <button type="submit">Find Route</button>
    </form>
</div>
```

```
</body>
</html>
```

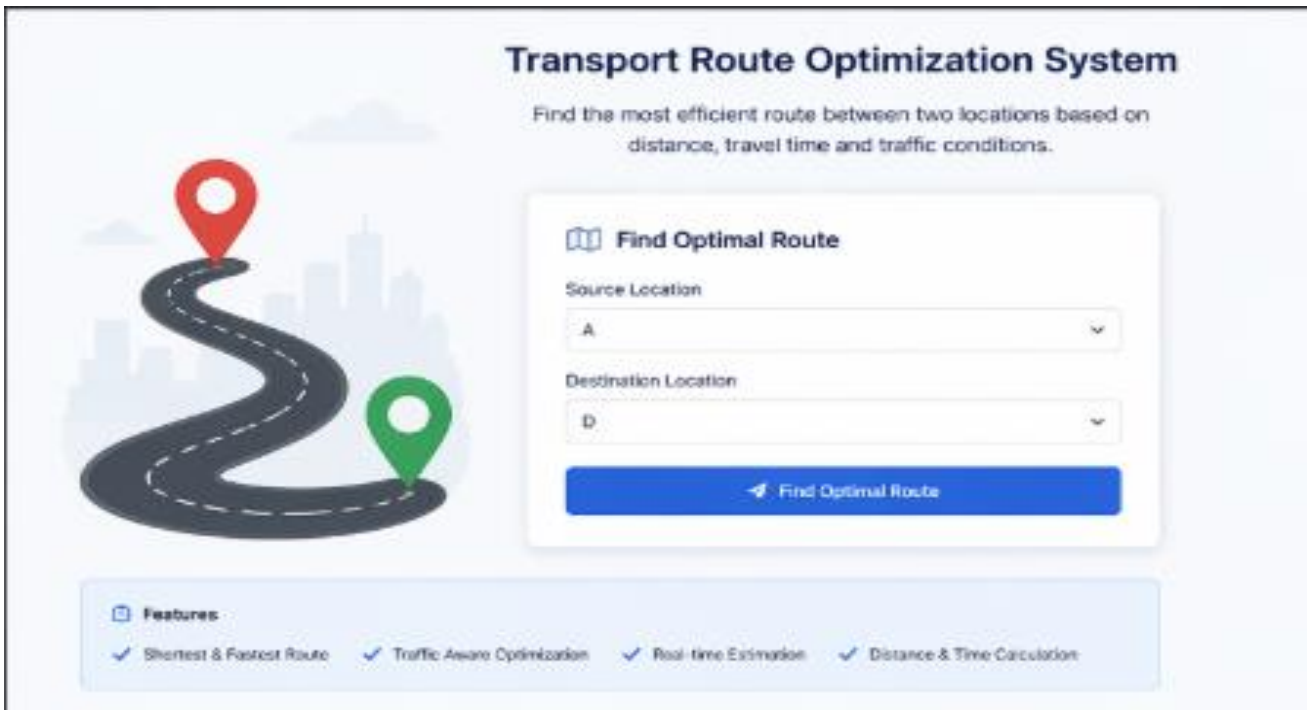
6. RESULT PAGE (templates/result.html)

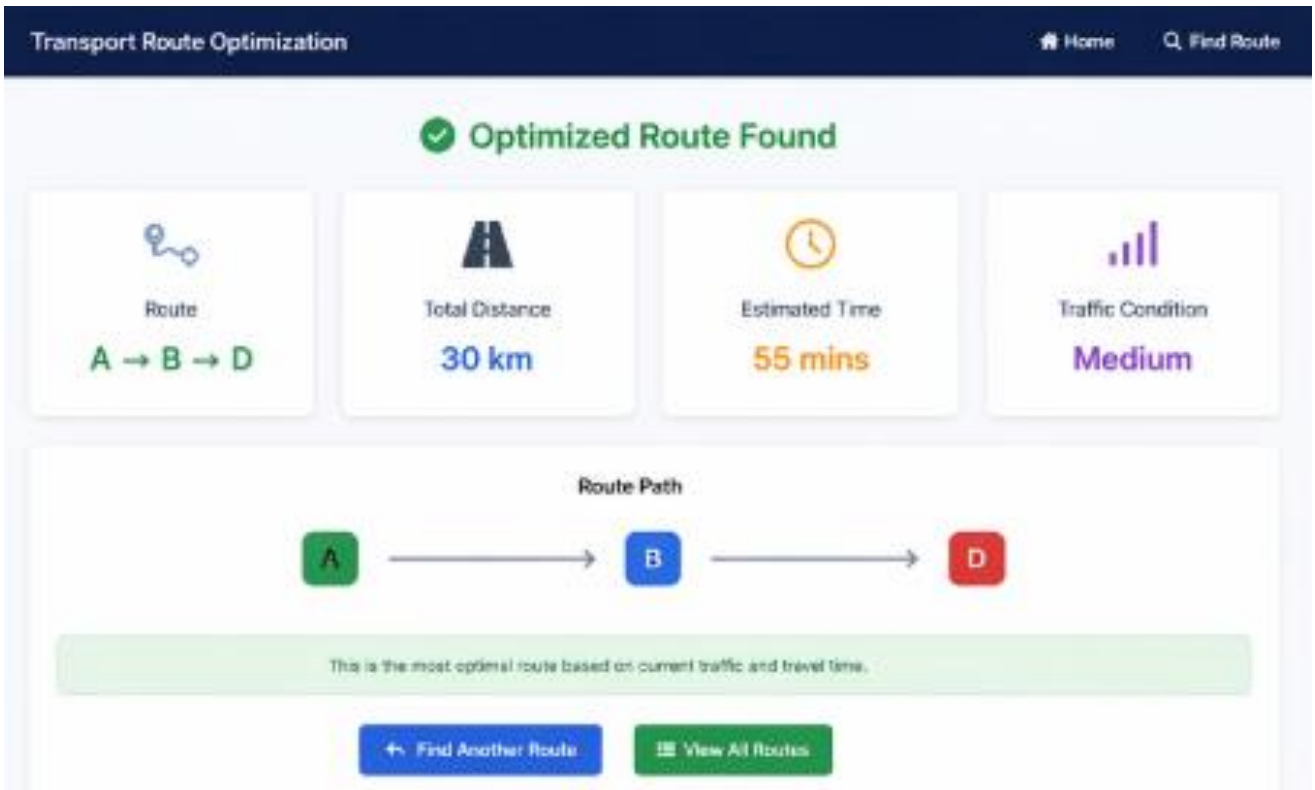
```
<!DOCTYPE html>
<html>
<head>
    <title>Result</title>
    <link rel="stylesheet" href="/static/style.css">
</head>
<body>

<div class="container">
    <h1>Optimized Route</h1>

    {% if error %}
        <h3>{{ error }}</h3>
    {% else %}
        <p><strong>Route:</strong> {{ path }}</p>
        <p><strong>Total Distance:</strong> {{ distance }} km</p>
        <p><strong>Estimated Time:</strong> {{ time }} mins</p>
    {% endif %}

    <a href="/">Go Back</a>
</div>

</body>
</html>
```

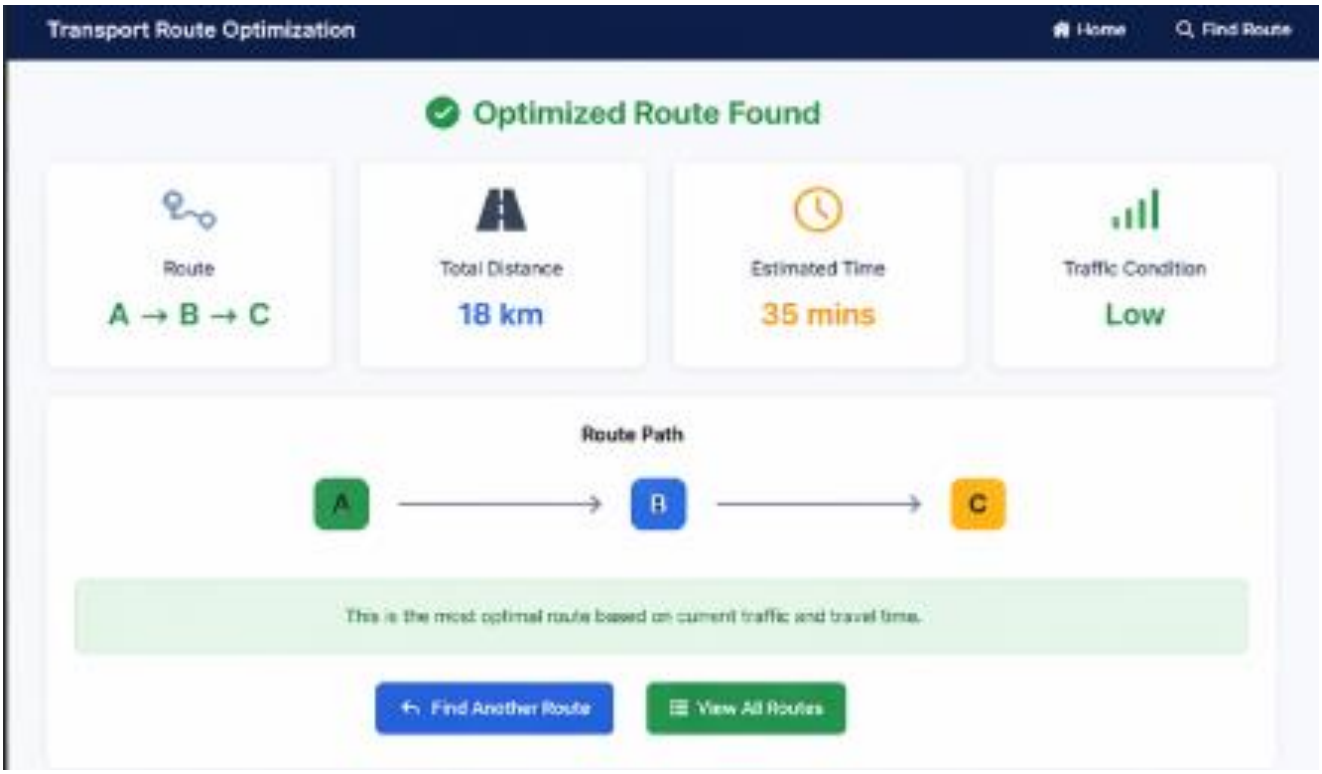
7. CSS (static/style.css)

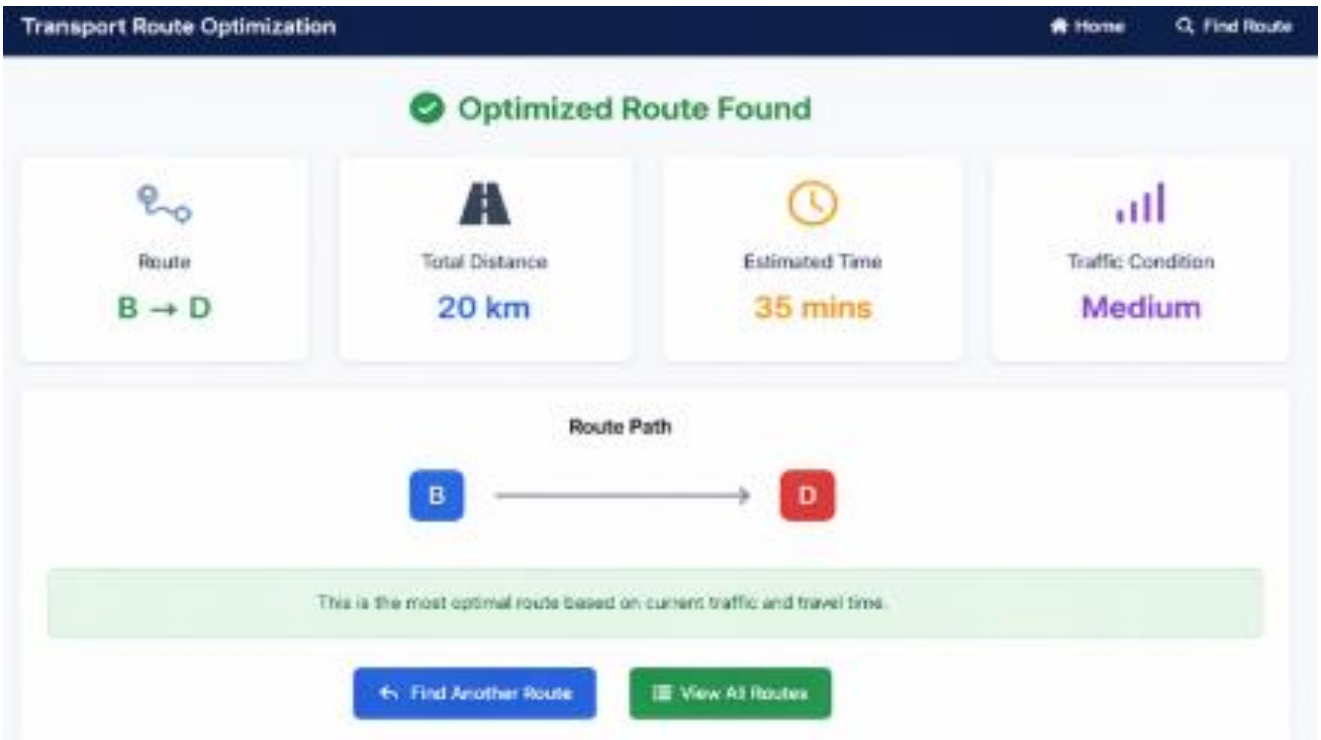
```
body {
    font-family: Arial;
    background-color: #f4f4f4;
}

.container {
    width: 400px;
    margin: 100px auto;
    padding: 20px;
    background: white;
    text-align: center;
    border-radius: 10px;

input, button {
    width: 90%;
    padding: 10px;
    margin: 10px;
}

button {
    background: #007BFF;
    color: white;
    border: none
```

OUTPUTS


Transport Route Optimization System
Find the most efficient route between two locations based on distance, travel time and traffic conditions.
Find Optimal Route
Source Location
A
Destination Location
D
Find Optimal Route
Features
Shortest & Fastest Route
Traffic Aware Optimization
Real-time Estimation
Distance & Time Calculation


Transport Route Optimization
Home
Find Route
Optimized Route Found
Route
A → B → D
Total Distance
30 km
Estimated Time
55 mins
Traffic Condition
Medium
Route Path
A
B
D
This is the most optimal route based on current traffic and travel time.
Find Another Route
View All Routes


Transport Route Optimization
Home
Find Route
Optimized Route Found
Route
A → B → C
Total Distance
18 km
Estimated Time
35 mins
Traffic Condition
Low
Route Path
A
B
C
This is the most optimal route based on current traffic and travel time.
Find Another Route
View All Routes


Transport Route Optimization
Home
Find Route
Optimized Route Found
Route
B → D
Total Distance
20 km
Estimated Time
35 mins
Traffic Condition
Medium
Route Path
B
D
This is the most optimal route based on current traffic and travel time.
Find Another Route
View All Routes

RESULTS AND FUTURE ENHANCEMENT

RESULTS

The Transport Route Optimization Analytics System was successfully developed and tested using travel and traffic datasets. The system effectively analyzed important factors such as distance, travel time, and traffic conditions to determine the most efficient routes between locations.

The data preprocessing stage successfully handled missing values, duplicate records, and inconsistent traffic data, resulting in a clean and structured dataset for analysis.

Exploratory Data Analysis revealed clear patterns in traffic behavior. It was observed that routes with shorter distances do not always result in minimum travel time due to traffic congestion. High traffic density significantly increases travel time, especially during peak hours.

The implementation of route optimization algorithms such as Dijkstra's Algorithm provided accurate results in finding the shortest and fastest routes. The system successfully calculated optimal paths based on weighted factors like travel time and traffic levels.

The generated graphs and charts such as Traffic Density Analysis, Distance vs Travel Time, and Traffic Heatmap provided meaningful insights into traffic patterns and route efficiency.

The Flask-based web application successfully allowed users to input source and destination locations and receive instant optimized route results, including route path, total distance, and estimated travel time.

Overall, the project demonstrates how data analytics and optimization algorithms can improve transportation efficiency, reduce travel time, and support better route planning.

FUTURE ENHANCEMENTS

- Integrate real-time traffic data using APIs such as Google Maps API
- Implement GPS-based live route tracking system
- Use Machine Learning models to predict future traffic conditions
- Develop a mobile application for easy user access
- Add interactive maps for route visualization
- Implement dynamic rerouting based on live traffic updates
- Include weather conditions for better route prediction
- Build dashboard using Power BI or Tableau for traffic analytics
- Support multiple routes comparison feature
- Improve system scalability for large city datasets
- Add voice navigation and alerts for drivers

- Integrate with smart city infrastructure for advanced traffic managemen

GitHub Link of the project along with the report	https://github.com/santhosh-kr714/DA-MINI-PROJECT-REPORT

REFERENCES

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- Kaggle, Traffic and Transportation Datasets. Available at: https://www.kaggle.com/
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